

E.2 SFT on Reasoning Traces from S1K

Model: LLaDA+SFT

Generation Sequence Length: 512

Question: A 76-star flag has three rows of 8 stars two rows of 6 stars and the rest are 5-star rows. How many rows of 5 stars are there altogether on the flag?

<reasoning> To determine the number of rows of 5 stars on the 76-star flag, we need to follow these steps:

1. Calculate the total number of stars in the three rows of 8 stars. 2. Calculate the total number of stars in the two rows of 6 stars. 3. Subtract the total number of stars in the first two rows from the total number of stars on the flag to find the number of stars in the remaining rows. 4. Divide the number of stars in the remaining rows by 5 to find the number of 5-star rows.

Let's go through each step in detail:

1. Calculate the number of stars in the three rows of 8 stars:

$$3 \times 8 = 24$$

So, there are 24 stars in the three rows of 8 stars.

2. Calculate the number of stars in the two rows of 6 stars:

$$2 \times 6 = 12$$

So, there are 12 stars in the two rows of 6 stars.

3. Subtract the total number of stars in the first two rows from the total number of stars on the flag:

$$76 - 24 - 12 = 40$$

So, there are 40 stars in the remaining rows.

4. Divide the number of stars in the remaining rows by 5 to find the number of 5-star rows:

$$\frac{40}{5} = 8$$

So, there are 8 rows of 5 stars.

However, we need to check if the number of rows is 8, as if there are 8 rows of 5 stars, the total number of stars would be:

$$8 \times 5 = 40$$

This matches the number of stars in the remaining rows, which is 40. Therefore, the number of rows of 5 stars is indeed 8.

Thus, the number of rows of 5 stars is:

8

< /reasoning><answer>8< /answer>

Model: LLaDA+SFT

Generation Sequence Length: 512

Question: A bakery produces 60 loaves of bread each day. Two-thirds of the loaves are sold in the morning and half of what is left is sold equally in the afternoon and evening. How many loaves of bread are sold in the afternoon?

<reasoning>To determine the number of loaves of bread sold in the afternoon, we need to follow these steps:

1. Calculate the number of loaves sold in the morning. 2. Determine the number of loaves remaining after the morning sales. 3. Calculate the number of loaves sold in the afternoon and evening. 4. Identify the number of loaves sold in the afternoon.

First. Calculate the number of loaves sold in the morning:

$$\text{Loaves sold in the morning} = \frac{2}{3} \times 60 = 40$$

Next. Determine the number of loaves remaining after the morning sales:

$$\text{Loaves remaining after morning sales} = 60 - 40 = 20$$

Then. Calculate the number of loaves sold in the afternoon and evening:

$$\text{Loaves sold in the afternoon and evening} = \frac{1}{2} \times 20 = 10$$

Since, half of the remaining loaves are sold equally in the afternoon and evening, we need to find the number of loaves sold in the afternoon..

Finally the number of loaves sold in the afternoon is:

$$\text{Loaves sold in the afternoon} = \frac{10}{2} = 5$$

However, this seems incorrect because the problem states that half of the remaining left is sold equally in the afternoon and evening, which means the should be:

$$\text{Loaves sold in the afternoon} = \frac{1}{2} \times 20 = 10$$

But since the problem states that half of the remaining left is sold equally in the afternoon and evening, the correct calculation should be:

$$\text{Loaves sold in the afternoon} = \frac{1}{2} \times 20 = 10$$

However, since the problem states that half of the remaining left is sold equally in the afternoon and evening, the correct calculation should be:

$$\text{Loaves sold in the afternoon} = \frac{1}{2} \times 20 = 10$$

< /reasoning> <answer> 10 < /answer>

E.3 *diffu*-GRPO

Model: LLaDA+*diffu*-GRPO

Generation Sequence Length: 512

Question: A 76-star flag has three rows of 8 stars two rows of 6 stars and the rest are 5-star rows. How many rows of 5 stars are there altogether on the flag?

<reasoning> To determine the number of rows of 5 stars on the flag, we first need to calculate the total number of stars in the rows with 8 stars and 6 stars.

1. There are number of rows with 8 stars: $3 \times 8 = 24$ stars.

2. There are number of rows with 6 stars: $2 \times 6 = 12$ stars.

3. The these rows rows together have $24 + 12 = 36$ stars.

Next, we need the total number of stars on the flag, which is 76. We subtract the number of stars in the rows with 8 stars and 6 stars from the total number of stars:

$$76 - 36 = 40$$

So, there are 40 stars in the rows with 5 stars. To find the number of rows of 5 stars, we divide the number of stars in these rows by the number of stars per row:

$$\frac{40}{5} = 8$$

Therefore, there are 8 rows of 5 stars on the flag.

< /reasoning><answer> 8 < /answer>

Model: LLaDA+*diffu*-GRPO

Generation Sequence Length: 512

Question: A bakery produces 60 loaves of bread each day. Two-thirds of the loaves are sold in the morning and half of what is left is sold equally in the afternoon and evening. How many loaves of bread are sold in the afternoon?

The bakery produces 60 loaves of bread each day. Two-thirds of the loaves are sold in the morning, so we calculate $\frac{2}{3} \times 60 = 40$ loaves sold in the morning. After the morning sales, there are $60 - 40 = 20$ loaves left. Half of these remaining loaves are sold equally in the afternoon and evening, so we calculate $\frac{1}{2} \times 20 = 10$ loaves sold in the afternoon.

Therefore< /reasoning><answer> 10 < /answer>